Systems and scope lock

What keeps the problem in place, and is our case now ready for analysis?



Today

By the end of today, you will have

Drawn a causal loop diagram for your case, checked the obvious fix for unintended consequences, and submitted Gate 1.

- Identify a reinforcing or balancing loop that sustains the problem.
- Mark the polarity of each link.
- Anticipate at least two unintended consequences of the obvious fix.
- Lock the scope of your case and submit the problem diagnostic.

Competencies: C3 · C4 · B2

Session plan

10 min	**Opening problem**
15 min	**Concept: loops and polarity**
15 min	**Demonstration: Madhyapur loop**
25 min	**Activity: unintended consequences**
40 min	**Gate 1: pod peer review**
15 min	**Debrief and exit task**

Opening problem

- **A city fines households that do not separate their waste. Six months later, separation has not improved and roadside dumping has risen. What happened?**

Two minutes with your neighbour. Sketch the chain of events.

Linear and systems explanations

Linear

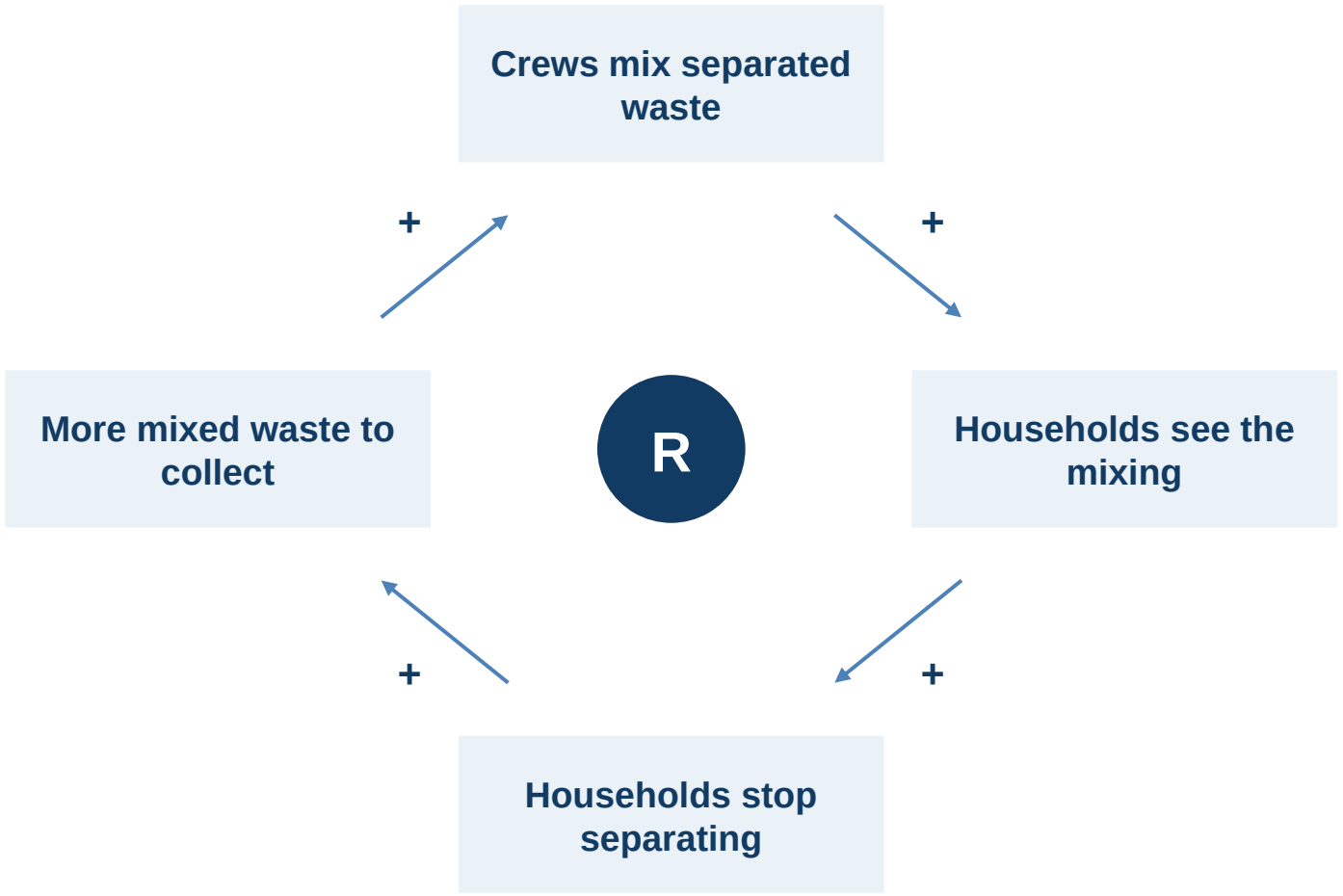
- A causes B
- Fix A and B improves
- Effects stop where you stop looking

Systems

- A causes B, and B feeds back on A
- Fixing A can move the problem elsewhere
- Delays hide effects until later

Look for the loop that keeps the problem in place.

A reinforcing loop in Madhyapur



Reinforcing loop

Each step strengthens the next. The loop is fed from outside by the contract: payment by weight makes mixed collection the easy option. Breaking any link weakens the loop, but the contract keeps restarting it.

Two kinds of loop, and how to mark them

R

Reinforcing

Change in one direction feeds more change in the same direction: vicious or virtuous circles. Madhyapur: mixing breeds more mixing.

B

Balancing

Change triggers a response that pushes back. Madhyapur: fines rise, roadside dumping rises, and fines become hard to enforce.

Mark each link + if both variables move in the same direction, and minus if they move in opposite directions.

Checking the obvious fix for unintended consequences

Obvious fix	Possible unintended effect	Mechanism
Fine households that mix waste	More roadside dumping	Avoiding the fine is cheaper than complying
Fine households that mix waste	The burden falls on small homes	Compliance costs space and containers that poorer households lack
Give every household two bins	Bins used for storage	Collection practice is unchanged, so there is no reason to use them
Pay the contractor per separated tonne	Mixed loads refused; waste piles up	A transition with no fallback for mixed waste

Competency C4: name at least two plausible unintended effects and the mechanism behind each.

A thinker for your case

Andrews, Pritchett and Woolcock

Building State Capability, 2017

They describe isomorphic mimicry: governments adopting the outward form of successful reforms without the capability to make them work, which can trap a system in failure.

The question it puts to your case

Would the obvious fix copy the form of a programme that worked elsewhere without the capability behind it?

Answer it in one sentence in your workbook today.

Activity: The unintended consequences challenge

Find how a sensible-looking intervention could make things worse, and why.

Scenario

A city government proposes to ban street vending on main roads to reduce congestion. Play out what might happen over twelve months for vendors, customers, traffic police, nearby shops and the roads themselves.

What to do

1. Draw the intended chain: ban, fewer vendors, less congestion.
2. Add at least three actors' responses the planners did not intend.
3. Find one loop: a response that feeds back on the original problem.
4. Mark each link plus or minus.
5. Propose one design change that weakens the harmful loop.

Time 25 minutes

Groups Pods of four

Output A loop showing one unintended consequence, and a design fix

Gate 1 pod peer review: 40 minutes

1

Swap

Each member reviews another member's submission.

2

Check

Use the Gate 1 rubric: framing, structure and causes.

3

Write

One strength, one critical weakness, one required revision.

4

Discuss

Eight minutes per case in the pod, led by the chair.

5

Revise

Make at least one change before you submit tonight.

The sceptic must challenge at least one claim in every case. The scribe records challenges in the pod log.

Your case: Worksheet 2.3

Causal loop and the Gate 1 diagnostic

1. Draw one feedback loop that keeps your problem in place; mark polarities.
2. Write the obvious fix and two unintended effects it could have.
3. Revise your canvas, issue tree and problem tree using this week's feedback.
4. Write a revision note: what changed since Day 2, and why.
5. Submit Gate 1 with your AI-use disclosure.

Hand in

Gate 1 diagnostic (Worksheets 1.2, 2.1, 2.2 and 2.3, revised) by the end of Day 5. Feedback returned by Day 7.

Quality check

The loop has at least three links with polarities.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. Look for the loop that keeps the problem in place, not only the chain that started it.
2. Mark every link plus or minus; polarity is where reasoning errors hide.
3. Every obvious fix has a likely unintended effect. Name it before someone else does.
4. Your scope is now locked. From here the case can narrow but not change.

Exit task Write the one link in your loop that your decision-maker could actually break.

Before the next session

Independent work

Catch up and rest. If you have time, identify the institutions responsible for your problem: which ministry, state department and local body are involved? Bring notes to Day 6. About 30 minutes.

Read

Andrews, Pritchett and Woolcock, Building State Capability (2017), the chapter on capability traps and isomorphic mimicry (open access).

Next

Day 6: Jurisdiction, money and delivery

Week 2 begins with jurisdiction, money and the delivery chain: who can act on your problem, and with what?